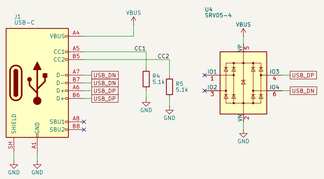
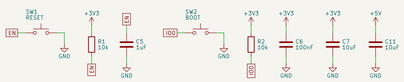
POWER IN
5V DC terminal or USB, ORed Schottky diodes

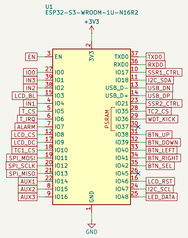
USB-C
native USB flashing + ESD



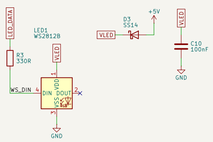
RESET / BOOT / DECOUPLING



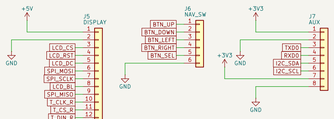
ESP32-S3-WROOM-1U-N16R2
GPIOs = firmware Kconfig defaults



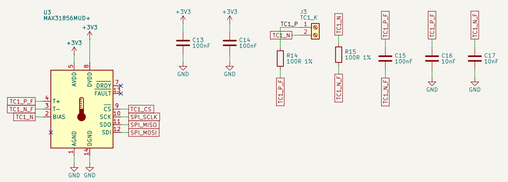
WS2812B STATUS LED
VDD dropped ~4.6V for 3.3V data margin



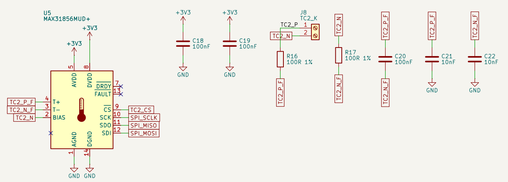
HEADERS
J5 display+touch (14-pin), J6 nav, J7 aux+I2C



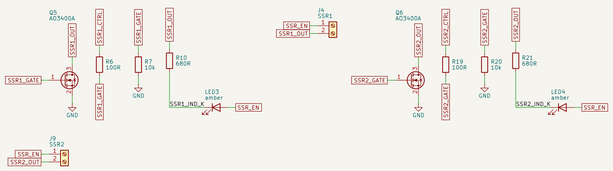
THERMOCOUPLE 1 (control) MAX31856
T- floats to J3, biased via BIAS



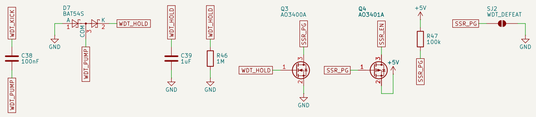
THERMOCOUPLE 2 (load) MAX31856
T- floats to J8, biased via BIAS



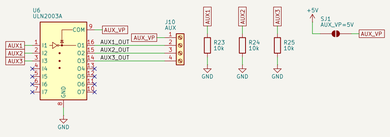
SSR DRIVE x2
low-side AO3400A; indicator LED across each terminal



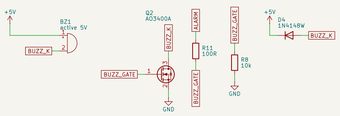
HARDWARE WATCHDOG
C38/D7/C39/R46 diode charge pump on WDT_KICK holds Q3 on; Q3 pulls SSR_PG down, turning on high-side Q4, which supplies SSR_EN — the +5V rail feeding BOTH SSR terminals and both indicators. R47 is the fail-safe pull-up.
SJ2 = bring-up defeat, REMOVE for service.



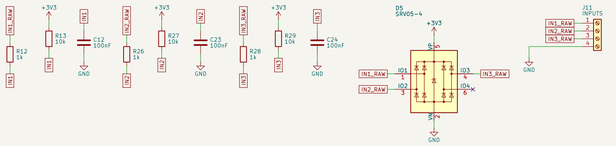
AUX OUTPUT BANK
ULN2003 vent/purge/spare; COM->AUX_VP, SJ1 links to +5V



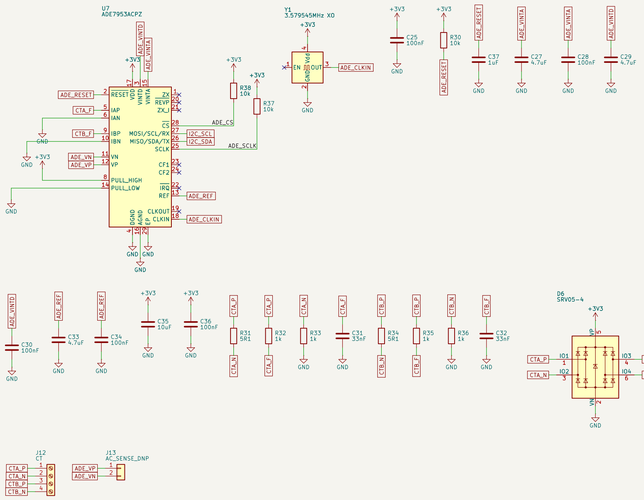
ALARM BUZZER



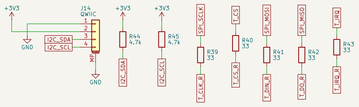
PROTECTED INPUTS x3 lid / gas-flow / spare
1k + 10k pull-up + 100nF each, SRV05-4 TVS



CT CURRENT SENSING ADE7953, I2C, current-only
no mains — VP/VN to DNP J13



TOUCH DAMPING + I2C EXPANSION
R39–R43 damp the shared SPI2 bus for the display module's XPT2046 (not on this board); J14 Qwiic and J7 5–8 (0.1 in) share the I2C bus, pulled up by R44/R45



TEST POINTS

MOUNTING / FIDUCIALS / POWER FLAGS
FID1–3 are bare-copper pick-and-place targets, on no net

NOTES — Bisque kiln controller, rev B

SSR terminals J4/J9: pin1 = SSR_EN (board +5V, watchdog-gated), pin2 = the switched low side. Boot-safe: R7/R20 hold each MOSFET gate down while the ESP32 pins are high-impedance, NOT opto-isolated — rev B tried that and reverted it: an opto only isolates if the SSR loop is powered off-board, and this board powers it.

TC1 terminal J3 / TC2 terminal J8: pin1 = K+, pin2 = K− — both float, biased near AGND only through each MAX31856's internal BIAS network. Ungrounded-junction probes required; two grounded-junction probes in one kiln would loop through it.

Nav switch J6 is panel-mounted; inputs use ESP32 pull-ups.

J11: IN1/IN2/IN3 (lid/gas-flow/spare) + GND — dry contact each channel to GND.

Display J5 (14-pin, ST7796S + XPT2046 touch module, LCDWIKI MSP4021): 1=+5V 2=GND 3=CS 4=RST 5=DC 6=SDI/MOSI 7=SCK 8=BL 9=SDO/MISO 10=T_CLK 11=T_CS 12=T_DIN 13=T_DO 14=T_IRQ. PIN 1 IS +5V, NOT 3V3 — the module regulates on board; do not wire a 3.3V-only panel to it. Logic is 3.3V; touch shares SPI2 with the LCD and both MAX31856s (R39–R43 damping).

J12: CT current inputs, CTA_P/CTA_N/CTB_P/CTB_N — one CT clamp per SSR zone.

J13: DNP SELV voltage sense header, NOT mains-rated, not fitted — do not wire to AC mains. Y1/C25/C26 load caps are an ASSUMED value (unverified C_L); see design.py comment. ADE7953 I2C address 0x38 collides with PCF8574A.

4-layer, 100 x 100 mm, JLCPCB standard process
ESP32-S3-WROOM-1U-N16R2 + 2x MAX31856 + dual SSR + ADE7953
Bisque project
Sheet: /
File: bisque-controller.kicad_sch
Title: Bisque Kiln Controller
Size: A1 | Date: 2026-07-20 | **Rev: B**
KiCad E.D.A. 10.0.5 | Id: 1/1